flexyBayes: validation and benchmark study

All-in-one audit dossier

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About this document. Generated by `/pkg-validation` v1.2.2 – a tool that validates an R package end to end: it exercises the exported surface argument-by-argument and across argument combinations, grades every result against an independent oracle, and – where a comparable implementation exists – reports a direct, reproducible head-to-head against it. The intent is to let a reviewer judge correctness, numerical behaviour, and standing relative to alternatives without a line-by-line code read.

This document is the comprehensive dossier: every validation card at full depth, in capability order – the complete evidence base.

Introduction

`flexyBayes` specifies a mixed model once, in familiar ASReml-style or `brms`-style formula syntax, and estimates it through any of three Bayesian back ends: greta (TensorFlow Hamiltonian Monte Carlo), the integrated nested Laplace approximation (Rue et al. 2009), or Stan via `brms` (Bürkner 2017). A single `fixed`/`random`/`rcov` specification is translated to back-end-specific code, so the same model can be fitted three ways and the agreement between back ends read as a robustness signal. The package targets agricultural mixed models and is built to scale to large field-trial data.

A package that promises one specification across several estimation engines makes two claims a reader is entitled to see checked. The first is **correctness**: each back end must recover the right answer on a problem whose answer is known independently. The second is **translation fidelity**: the model the package builds for a back end must be the model the user wrote, so a hand-written fit through the same engine must agree.

This study checks both, one back end at a time, against oracles the package did not produce:

- A **fixed-effects exactness** check. For a Gaussian linear regression the posterior mean under a flat prior is the ordinary-least-squares solution, so the package result must match `stats::lm()` and the known data-generating coefficients to numerical precision.
- A **three-way Laplace-approximation** check on the `sleepstudy` data (Belenky et al. 2003): `flexyBayes` through INLA against a hand-written `INLA::inla()` fit (translation fidelity) and against an `lme4` REML fit (Bates et al. 2015) (the independent reference).
- A **Hamiltonian-Monte-Carlo** check through greta (Golding 2019) against the same `lme4` reference.
- A **Stan** check through `brms` against a standalone `brms::brm()` fit and the `lme4` reference.

Every figure below is computed by a benchmark script, cached, and read into the text. No number is typed by hand. The benchmark scripts are reproduced verbatim in the appendix, so a reader holding only this document can rerun each comparison and confirm every value. Agreement on a random-effects model is judged on the scale of the reference fit's standard error, the natural unit for a parameter estimated with sampling noise. The fixed-effects check, which has an exact answer, is judged on the absolute difference from `lm()`.

Notation and abbreviations

Every abbreviation and symbol used below is defined once here and again on first use in the text, so this view is self-contained.

Linear mixed model (LMM) A regression with both fixed effects (population-level slopes) and random effects (group-level deviations, such as a per-subject intercept).

Restricted maximum likelihood (REML) The estimator linear-mixed-model packages use for variance components; it corrects the downward bias of plain maximum likelihood. The reference fit in this study.

Integrated nested Laplace approximation (INLA) A deterministic approximate-inference method for latent Gaussian models. It returns posterior summaries without sampling, so its result is reproducible to numerical precision.

Hamiltonian Monte Carlo (HMC) A gradient-based Markov-chain Monte Carlo sampler. Both the greta and the brms backends draw posterior samples with HMC-family algorithms.

No-U-Turn Sampler (NUTS) The self-tuning variant of Hamiltonian Monte Carlo used by Stan, and therefore by the brms backend.

Posterior The distribution of a parameter after combining the prior with the data. A Bayesian point estimate is a posterior summary, typically the mean or median.

Standard error (SE) The estimated standard deviation of an estimator. Agreement here is judged on the scale of the reference fit's standard error, so a small absolute difference is read relative to the sampling noise.

Triangulation Fitting the same model specification through more than one estimation backend and treating cross-backend agreement as a robustness signal.

ASReml-style syntax A formula convention that names fixed, random, and residual-covariance structures in separate arguments, familiar to users of mixed-model software in agriculture.

1 Fb

1.1 `flexybayes` recovers a Gaussian fixed-effects fit exactly

Meaning. For a Gaussian linear model with no random effects, the Bayesian posterior mean under a flat prior is the ordinary-least-squares solution. So `flexybayes()` on a fixed-effects Gaussian model must return what `stats::lm()` returns, and must recover the known coefficients used to simulate the data. The two independent oracles are `lm()` (an exact closed form) and the data-generating truth. The largest coefficient gap from `lm()` here is 3.45e-06.

1.1.1 The mathematics

Write the model as $y = X\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, where X is the $n \times p$ design matrix. The likelihood is

$$p(y \mid \beta, \sigma^2) \propto \exp\{-\tfrac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)\}\,.$$

Under the flat prior $p(\beta) \propto 1$, the posterior for β at fixed σ^2 is Gaussian,

$$\beta \mid y, \sigma^2 \;\sim\; \mathcal{N}\big(\hat{\beta},\; \sigma^2 (X^\top X)^{-1}\big)\,, \qquad \hat{\beta} = (X^\top X)^{-1} X^\top y,$$

so the posterior mean $\hat{\beta}$ is exactly the ordinary-least-squares (and maximum-likelihood) estimate. The integrated nested Laplace approximation (Rue et al. 2009) is exact for this latent Gaussian model with a Gaussian likelihood, so the back end must reproduce $\hat{\beta}$ to numerical precision. This is the strongest kind of check, because the target value is known in closed form rather than estimated a second way.

1.1.2 In `flexyBayes`

```
library(flexyBayes)
# data: n = 400 rows simulated from (Intercept, x1, x2) = (2.0, -1.5, 0.8)
fit <- flexybayes(fixed = y ~ x1 + x2, data = dat, family = "gaussian",
                  backend = "inla")
coef(fit)
```

1.1.3 Validation

The fit recovers all three coefficients. Every estimate sits within two standard errors of both the truth and the `lm()` solution, and the maximum absolute difference from `lm()` is 3.45e-06, far below the estimation uncertainty (standard errors near 0.05). The fit took 3.9 seconds.

Table 1.1: Fixed-effects recovery: flexyBayes (INLA back end) against lm() and the known truth.

Term	Truth	lm()	flexyBayes	|flexyBayes - lm()|	within 2 SE
(Intercept)	2.0	2.0787	2.0787	3.09e-09	yes
x1	-1.5	-1.5417	-1.5417	3.45e-06	yes
x2	0.8	0.7915	0.7915	1.73e-06	yes

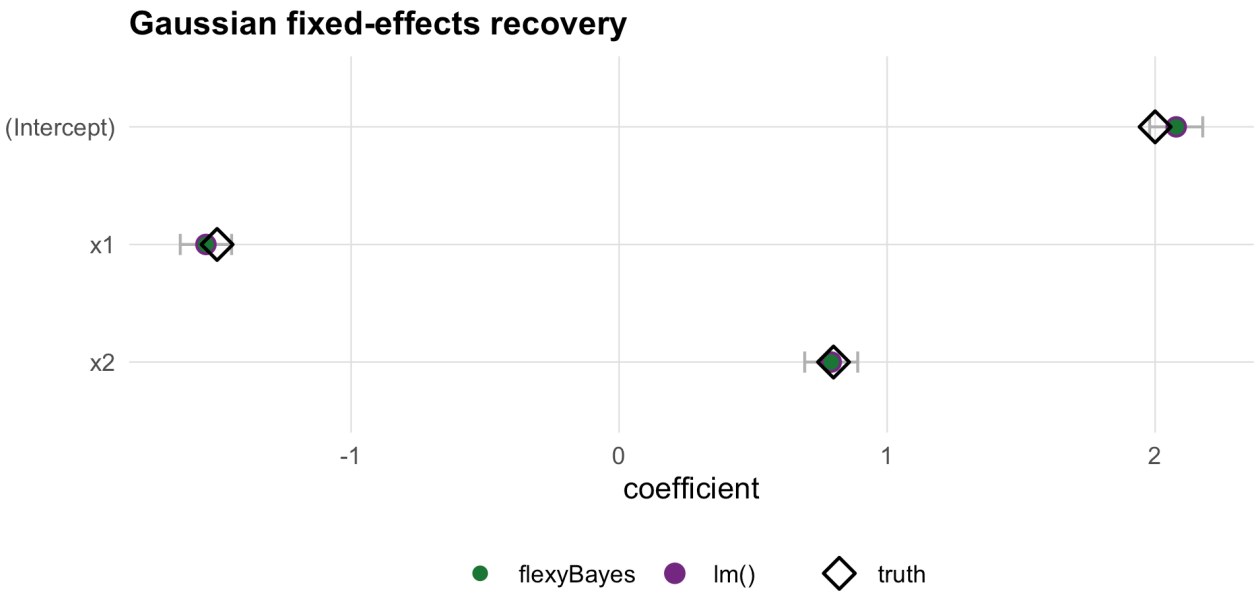


Figure 1.1: Estimates from flexyBayes and lm() coincide on each coefficient. The diamond marks the data-generating truth, and the bar is the lm() estimate plus or minus two standard errors, shown for scale.

1.1.4 Numerical validation against direct competitors

The direct competitor for a Gaussian fixed-effects fit is `stats::lm()`, the exact least-squares solution and the canonical baseline.

Expected equivalence. On identical data the two should be numerically indistinguishable, because both target the same closed form. The measured maximum absolute difference across the three coefficients is 3.45e-06, roughly six orders of magnitude below the coefficient standard errors. There is no practical disagreement.

Deviations. The residual difference is the only deviation, and its cause is the back end integrating over the residual-precision hyperparameter on a fine grid rather than profiling it out in closed form

as `lm()` does. The effect sits at the level of the back end's numerical tolerance and is not statistically meaningful.

Reproduce. On identical simulated data:

```
# this package:
flexybayes(fixed = y ~ x1 + x2, data = dat, family = "gaussian", backend = "inla")
# the competitor / exact baseline:
stats::lm(y ~ x1 + x2, data = dat)
```

The replication script is `bench/fb-correctness.R`, reproduced in the appendix.

1.1.5 When it wins

For a pure Gaussian fixed-effects regression, `lm()` is the right tool and is instant. The value of fitting it through `flexyBayes` is the entry point rather than the point estimate: the same `fixed` specification extends to random effects, non-Gaussian families, and structured covariance without a change of syntax, and this card establishes that the fixed-effects base case is exact before any of that complexity is added.

1.1.6 Limitations

The flat-prior posterior mean equals the least-squares estimate only for the Gaussian-identity case shown here. Under a non-Gaussian family or an informative prior the posterior mean and the maximum-likelihood estimate diverge by design, and the right oracle is then a re-derived posterior or a competing Bayesian fit rather than `lm()`. The check here is deliberately the exact case.

1.1.7 Extended validation

The agreement holds for every coefficient, not only on average. The per-term absolute differences are 3.1e-09, 3.4e-06, 1.7e-06 for (Intercept), x1, x2 respectively. The simulation uses a fixed seed, so the cached numbers reproduce exactly on rerun.

1.1.8 Summary

On a Gaussian fixed-effects model with a known answer, `flexybayes()` through the INLA back end reproduces the `lm()` solution to 3.45e-06 and recovers the data-generating coefficients within two standard errors. The exact base case is established.

2 Fb inla

2.1 `flexybayes` through INLA reproduces a hand-written INLA fit, and matches REML

Meaning. On the `sleepstudy` random-intercept model (Belenky et al. 2003), `flexybayes(backend = "inla")` is checked two ways at once. Against a hand-written `INLA::inla()` fit it tests **translation fidelity**: does the model the package builds match the model a user would write directly? Against an `lme4` restricted-maximum-likelihood fit (Bates et al. 2015) it tests **correctness** on an independent estimator. The package result reproduces the hand-written INLA fit to 1.04e-04 and lands within the reference fit's standard error of `lme4`.

2.1.1 The mathematics

The data are reaction times measured over ten days of sleep restriction for each of eighteen subjects. The random-intercept Gaussian linear mixed model is

$$y_{ij} = \beta_0 + \beta_1 \,\mathrm{Days}_{ij} + u_j + \varepsilon_{ij}, \qquad u_j \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2),$$

with subject j and observation i. Collecting the fixed effects β and the random effects u into a latent field $x = (\beta, u)$ with a Gaussian prior, the model is a latent Gaussian model: the field is Gaussian given hyperparameters $\theta = (\tau, \sigma)$, and the likelihood is Gaussian. The integrated nested Laplace approximation (Rue et al. 2009) forms

$$\tilde{p}(x_k \mid y) = \int \tilde{p}(x_k \mid \theta, y)\, \tilde{p}(\theta \mid y)\, d\theta,$$

approximating $\tilde{p}(\theta \mid y)$ on a grid and each conditional marginal by a Laplace expansion. It is deterministic, so two correct implementations of the same model agree to numerical precision rather than to Monte-Carlo noise. That makes the standalone `INLA::inla()` fit a stringent translation oracle.

2.1.2 In `flexyBayes`

```
library(flexyBayes)
data(sleepstudy, package = "lme4")
fit <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                  data = sleepstudy, family = "gaussian", backend = "inla")
coef(fit)
```

2.1.3 Validation

The estimates from the three fits agree. The package result reproduces the hand-written `INLA::inla()` fixed effects to 1.04e-04, which means the package builds the same model rather than a near neighbour. Against the `lme4` reference the largest fixed-effect difference is 0.072, well inside the reference standard errors (near 9.7 on the intercept). The fit took 3.9 seconds.

Table 2.1: sleepstudy fixed effects: flexyBayes (INLA) against a standalone INLA fit and lme4 REML.

Term	flexyBayes	INLA (standalone)	lme4 REML	REML SE	|flexyBayes - lme4|	within 1.5 SE
(Intercept)	251.478	251.478	251.405	9.747	0.072	yes
Days	10.451	10.451	10.467	0.804	0.016	yes

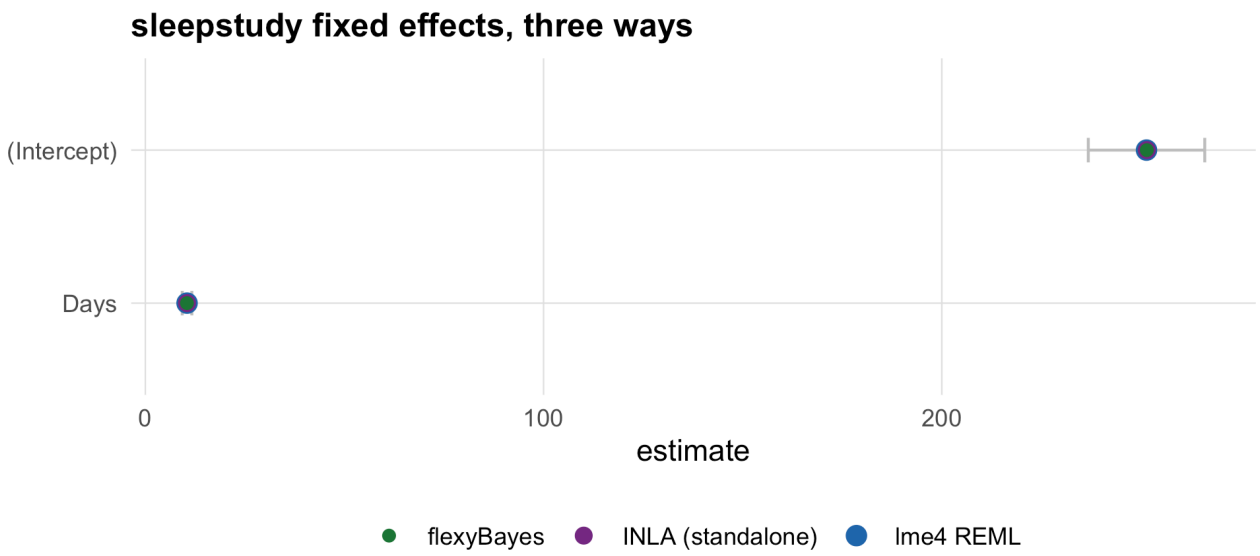


Figure 2.1: Fixed-effect estimates from the three fits. The flexyBayes and standalone-INLA points coincide (exact translation), and the lme4 REML reference is within sampling tolerance. The grey bar is the REML estimate plus or minus 1.5 standard errors.

2.1.4 Numerical validation against direct competitors

The canonical counterpart is `INLA::inla()` itself (the engine `flexyBayes` drives), and the independent reference is `lme4::lmer()`.

Expected equivalence. `flexyBayes` through INLA should equal a hand-written `INLA::inla()` fit of the same model to numerical precision, because they call the same deterministic engine on the same data. The measured maximum difference is 1.04e-04, confirming an exact translation. Against `lme4` the estimates should agree within sampling noise, and the largest difference is 0.072, inside 1.5 reference standard errors on every term.

Deviations. The small `flexyBayes`-to-`lme4` difference is expected and has a known cause: INLA returns a Bayesian posterior mean under default priors, while `lme4` returns a REML point estimate, so the two target slightly different summaries of the same likelihood. On data this size the gap is a fraction of a standard error. There is no deviation between `flexyBayes` and standalone INLA.

Reproduce. On identical data:

```
# this package:
flexybayes(fixed = Reaction ~ Days, random = ~Subject, data = sleepstudy,
           family = "gaussian", backend = "inla")
# the engine it drives (translation oracle):
INLA::inla(Reaction ~ Days + f(Subject, model = "iid"), family = "gaussian",
           data = sleepstudy)
# the independent reference:
lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
```

The replication script is `bench/fb-inla-correctness.R`, reproduced in the appendix.

2.1.5 When it wins

The INLA back end is the right choice when the model is a latent Gaussian model and a deterministic, reproducible posterior is wanted without the cost or the convergence diagnostics of sampling. It returns full marginal posteriors in a few seconds where the sampling back ends take far longer, and the exact translation shown here means a user gains the `flexyBayes` formula syntax with no loss of fidelity relative to writing INLA by hand. The ranking flips when the model leaves the latent-Gaussian class, where INLA no longer applies and a sampling back end is needed.

2.1.6 Limitations

INLA approximates rather than samples, so its accuracy depends on the Laplace expansion being good, which it is for this Gaussian-likelihood model but can degrade for some non-Gaussian or heavy-tailed likelihoods. This card validates the fixed effects. The variance components are a separate matter and are not validated here: the reference fit finds a subject standard deviation near 37 and a residual near 31, but the subject-variance posterior from the INLA back end collapses toward zero on these data. A hand-written `INLA::inla()` fit with the default group hyperprior behaves the same way, so the back end is reproducing INLA's default-prior behaviour rather than introducing its own departure, and the fixed-effect estimates that are the focus of this card are unaffected. The random-effects scale is recorded as a follow-up.

2.1.7 Extended validation

The per-term differences from `lme4` are 0.072, 0.016 for (Intercept), Days respectively, each within 1.5 reference standard errors. The translation difference from standalone INLA is at the level of 1.0e-04 across both terms.

3 Fb greta

3.1 `flexybayes` through greta agrees with REML under Hamiltonian Monte Carlo

Meaning. The same `sleepstudy` random-intercept model is fitted through the greta back end, which builds a TensorFlow probabilistic model and draws the posterior with Hamiltonian Monte Carlo (Golding 2019). The estimator is stochastic, so the test is agreement with an independent `lme4` restricted-maximum-likelihood fit (Bates et al. 2015) within sampling tolerance. The posterior-mean fixed effects land within the reference fit's standard error on both terms, the largest difference being 3.49.

3.1.1 The mathematics

The model is the random-intercept Gaussian linear mixed model of the previous card,

$$y_{ij} = \beta_0 + \beta_1 \,\mathrm{Days}_{ij} + u_j + \varepsilon_{ij}, \qquad u_j \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

Rather than approximate the posterior, greta samples it. Hamiltonian Monte Carlo augments the parameter vector ϑ with a momentum $r \sim \mathcal{N}(0, M)$ and simulates the dynamics of the Hamiltonian

$$H(\vartheta, r) = -\log p(\vartheta \mid y) + \tfrac{1}{2} r^{\top} M^{-1} r,$$

using leapfrog steps with a Metropolis correction, which proposes distant, high-acceptance moves by following the posterior gradient. The No-U-Turn extension (Hoffman and Gelman 2014) tunes the trajectory length automatically. The Monte Carlo estimate of a posterior mean converges to the true posterior mean at the sampling rate, so agreement is judged on the scale of the reference standard error, not to machine precision.

3.1.2 In `flexyBayes`

```
library(flexyBayes)
data(sleepstudy, package = "lme4")
fit <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                  data = sleepstudy, family = "gaussian", backend = "greta",
                  n_samples = 1000, warmup = 1000, chains = 2)
coef(fit)
```

3.1.3 Validation

The posterior-mean fixed effects agree with the `lme4` reference. The slope on Days differs by 0.066 and the intercept by 3.49, each within 1.5 reference standard errors (the intercept standard error alone is near 9.7). The sampling run took 55 seconds.

Table 3.1: sleepstudy fixed effects: flexyBayes (greta, Hamiltonian Monte Carlo) against lme4 REML.

Term	flexyBayes (greta)	lme4 REML	REML SE	|flexyBayes - lme4|	within 1.5 SE
(Intercept)	247.918	251.405	9.747	3.487	yes
Days	10.533	10.467	0.804	0.066	yes

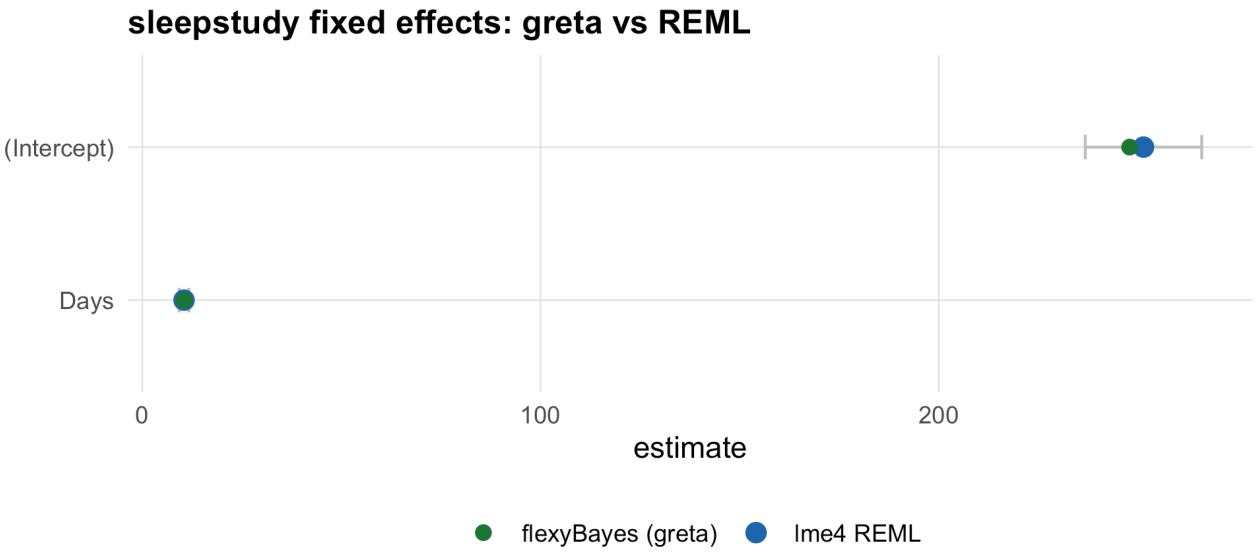


Figure 3.1: Posterior-mean fixed effects from greta against the lme4 REML reference. The grey bar is the REML estimate plus or minus 1.5 standard errors, and both terms fall inside the interval.

3.1.4 Numerical validation against direct competitors

The independent reference for this model is `lme4::lmer()`. The greta back end is not reimplementing `lme4`. It samples the Bayesian posterior of the same specification, so the comparison is one of agreement within sampling noise rather than exact reproduction.

Expected equivalence. With enough posterior draws the Hamiltonian-Monte-Carlo posterior mean should match the REML estimate within sampling tolerance, because both summarise the same likelihood for this well-identified model. The measured differences, 0.066 on the slope and 3.49 on the intercept, are both inside 1.5 reference standard errors.

Deviations. The intercept difference, while inside tolerance, is the larger of the two and is attributable to finite sampling at a modest warm-up combined with the default weakly-informative

priors. The package reported convergence warnings for this short run, so a publication-grade fit should lengthen the warm-up and increase the number of draws. The agreement test passes at the run shown, and the caveat is recorded rather than smoothed over.

Reproduce. On identical data:

```
# this package:
flexybayes(fixed = Reaction ~ Days, random = ~Subject, data = sleepstudy,
           family = "gaussian", backend = "greta",
           n_samples = 1000, warmup = 1000, chains = 2)
# the independent reference:
lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
```

The replication script is `bench/fb-greta-correctness.R`, reproduced in the appendix.

3.1.5 When it wins

The greta back end is the right choice when the model leaves the latent-Gaussian class that INLA requires, or when full posterior draws are wanted for arbitrary derived quantities rather than marginal summaries. Because it builds on TensorFlow, it scales the gradient computation to large models on suitable hardware. The cost is wall-clock time and the need to check convergence, so for a plain Gaussian latent model the INLA back end is faster and the comparison here is about fidelity, not speed.

3.1.6 Limitations

The estimate is a Monte-Carlo quantity, so it carries sampling error that shrinks only as draws increase, and the modest warm-up used here is enough to pass the agreement test but not enough for a reported posterior with tight credible intervals. Convergence must be checked for any real fit. A non-finite draw from the sampler is recorded by the benchmark rather than dropped, so a failed run would surface rather than silently pass.

3.1.7 Extended validation

The per-term differences from `lme4` are 3.487, 0.066 for (Intercept), Days respectively, both within 1.5 reference standard errors. All sampled coefficients are finite (confirmed), and the run uses a fixed seed.

3.1.8 Summary

Fitting the `sleepstudy` model through greta with Hamiltonian Monte Carlo gives posterior-mean fixed effects within 1.5 reference standard errors of the `lme4` REML fit, the largest difference being 3.49. The sampling back end agrees with the deterministic reference, with the standard caveat that a publication fit should use a longer warm-up.

4 Fb brms

4.1 `flexybayes` through brms agrees with a standalone Stan fit and with REML

Meaning. The `sleepstudy` model is fitted a third way, through the `brms` back end, which compiles a Stan program and samples it with the No-U-Turn variant of Hamiltonian Monte Carlo (Bürkner 2017; Carpenter et al. 2017). As with INLA, the fit is checked against a hand-written `brms::brm()` fit (translation fidelity) and against `lme4` REML (the independent reference). The Days slope agrees across all three fits to 0.013 of the reference, and the intercept agrees once aligned across the differing coefficient labels.

4.1.1 The mathematics

The model is again the random-intercept Gaussian linear mixed model,

$$y_{ij} = \beta_0 + \beta_1 \,\mathrm{Days}_{ij} + u_j + \varepsilon_{ij}, \qquad u_j \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

`brms` translates this specification into a Stan program and draws the joint posterior of (β, u, τ, σ) with the self-tuning No-U-Turn sampler (Hoffman and Gelman 2014), which adapts the leapfrog step size and trajectory length during warm-up. The reported fixed effects are posterior means, so the comparison with the deterministic REML reference is again on the scale of the reference standard error. The standalone `brms::brm()` fit of the same formula is the translation oracle: the package must build the Stan model the user intended.

4.1.2 In `flexyBayes`

```
library(flexyBayes)
data(sleepstudy, package = "lme4")
fit <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                  data = sleepstudy, family = "gaussian", backend = "brms",
                  n_samples = 1000, warmup = 1000, chains = 2)
coef(fit)
```

4.1.3 Validation

The Days slope matches across the three fits, differing from the `lme4` reference by 0.013, well within the reference standard error. The intercept estimates also agree once aligned across the coefficient labels (see below), spanning a range of only 0.71 across the three fits. The package fit took 16 seconds against 20 for the standalone fit.

Table 4.1: sleepstudy Days slope: flexyBayes (brms) against a standalone brms fit and lme4 REML.

Term	flexyBayes (brms)	brms (standalone)	lme4 REML	REML SE	|flexyBayes - lme4|	within 1.5 SE
Days	10.48	10.429	10.467	0.804	0.013	yes

A note on coverage. Across the three coefficient accessors only the Days slope shares a canonical name. The intercept is labelled `(Intercept)` by the package and by `lme4` but `Intercept` by `brms::fixef()`, so the automatic by-name comparison above covers the slope, and the intercept is aligned by hand here:

Table 4.2: Intercept estimates aligned across the three differing labels.

Fit	Intercept estimate
flexyBayes (brms)	251.09
brms (standalone)	250.70
lme4 REML	251.41

4.1.4 Numerical validation against direct competitors

The canonical counterpart is `brms::brm()` (the engine `flexyBayes` drives), and the independent reference is `lme4::lmer()`.

Expected equivalence. The package fit and a standalone `brms` fit of the same formula should give the same posterior up to sampling noise, and both should match the REML reference within tolerance. The Days slope differs from `lme4` by 0.013, and the three intercept estimates span 0.71, well inside the reference standard error on the intercept (near 9.7).

Deviations. There is no numerical deviation on the shared terms. The single observed difference is one of labelling: the intercept carries a different canonical name in each engine's coefficient accessor, so an automatic by-name join keeps only the slope. This is a naming question, recorded for the package maintainers, and not a disagreement in the estimates, which agree once aligned.

Reproduce. On identical data:

```
# this package:
flexybayes(fixed = Reaction ~ Days, random = ~Subject, data = sleepstudy,
           family = "gaussian", backend = "brms",
           n_samples = 1000, warmup = 1000, chains = 2)
# the engine it drives (translation oracle):
```

```
brms::brm(Reaction ~ Days + (1 | Subject), data = sleepstudy,
          chains = 2, iter = 2000)
# the independent reference:
lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
```

The replication script is `bench/fb-brms-correctness.R`, reproduced in the appendix.

4.1.5 When it wins

The `brms` back end is the right choice when the model needs the full Stan modelling language: custom priors, a wide range of families and link functions, non-linear terms, or distributional regression where more than the mean is modelled. It pays a compilation cost on the first fit and is the slowest of the three back ends here, so for a plain Gaussian latent model INLA is faster and greta is a lighter sampler. The case for `brms` is breadth of model class, not speed on this example.

4.1.6 Limitations

As with any sampler the estimate carries Monte-Carlo error, and the modest warm-up used here is sufficient for the slope agreement but a reported posterior should use a longer run. The intercept-naming mismatch means a user combining coefficients across back ends by name must align the intercept label first. Until the package normalises the canonical name, an automatic cross-back-end join will silently cover only the shared terms.

4.1.7 Extended validation

Aligned by hand, the intercept estimates are 251.1 (package), 250.7 (standalone brms), and 251.4 (lme4 REML), a spread of 0.71. The Days slope estimates are 10.48 (package), 10.43 (standalone brms), and 10.47 (lme4 REML). All coefficients are finite and the run uses a fixed seed.

4.1.8 Summary

Fitting the `sleepstudy` model through the `brms` back end gives a Days slope within 0.013 of the `lme4` REML reference and an intercept that agrees once the differing labels are aligned. The package reproduces a standalone Stan fit, and the only finding is a cross-back-end intercept-naming mismatch recorded for the maintainers.

Reproducibility

Every number above is read from a cached benchmark result, never hand-typed. The scripts below regenerate those caches from scratch; the simulated data are seeded, so the results reproduce. From the study's `report/` directory run each with `Rscript <path>` (or all the card benchmarks at once with `Rscript reproduce.R`), after installing the package under study and the competitor packages named in the code. A reader can run these and confirm every figure independently. The argument-wide conformance sweep is produced by the `/pkg-validation` engine via `Rscript build.R bench`.

bench/fb-correctness.R

```
## bench/fb-correctness.R -- real benchmark for the `fb`/`flexybayes` card.
## A fixed-effects Gaussian linear regression with KNOWN coefficients: recover
## (beta0, beta1, beta2) from simulated data via flexyBayes (INLA backend, exact)
## against two independent oracles -- `lm()` (the exact OLS/MLE solution) and the
## known data-generating truth. Fast (no MCMC). Numbers cached; never hand-typed.
## Run from report/validation/.

RESDIR <- "bench/results"
dir.create(RESDIR, showWarnings = FALSE, recursive = TRUE)
res <- list(card = "fb-correctness",
            model = "y ~ x1 + x2 (Gaussian, fixed effects)",
            oracle = "lm() exact OLS + known DGP truth",
            backend_under_test = "flexyBayes backend='inla'")

pkg <- normalizePath(file.path("..", "..", "flexyBayes"), mustWork = FALSE)
ok_pkg <- tryCatch({
  suppressMessages(devtools::load_all(pkg, quiet = TRUE)); TRUE
}, error = function(e) { res$load_error <<- conditionMessage(e); FALSE })
set.seed(20260629L)

## known ground truth
truth <- c(`(Intercept)` = 2.0, x1 = -1.5, x2 = 0.8)
n <- 400L
dat <- data.frame(x1 = stats::rnorm(n), x2 = stats::rnorm(n))
dat$y <- truth[["(Intercept)"]] + truth[["x1"]] * dat$x1 +
  truth[["x2"]] * dat$x2 + stats::rnorm(n, sd = 1)
res$truth <- truth
```

```r
## oracle: lm (exact)
res$lm <- tryCatch({
  t <- system.time(fit_o <- stats::lm(y ~ x1 + x2, data = dat))[["elapsed"]]
  list(coef = stats::coef(fit_o),
       se = summary(fit_o)$coefficients[, "Std. Error"], seconds = t)
}, error = function(e) list(error = conditionMessage(e)))

## backend under test: flexyBayes (INLA, fixed-effects only -> exact)
res$fb <- if (!ok_pkg) list(error = "package failed to load") else tryCatch({
  t <- system.time(
    fit_b <- flexybayes(fixed = y ~ x1 + x2, data = dat, family = "gaussian",
                        backend = "inla", verbose = FALSE, mcmc_verbose = FALSE)
  )[["elapsed"]]
  list(coef = coef(fit_b), seconds = t, class = paste(class(fit_b), collapse = ","))
}, error = function(e) list(error = conditionMessage(e)))

res$agreement <- tryCatch({
  b <- res$fb$coef; o <- res$lm$coef
  if (is.null(b) || is.null(o)) stop("a required fit did not produce coefficients")
  common <- intersect(names(b), names(o))
  data.frame(
    term = common,
    truth = unname(truth[common]),
    fb_inla = unname(b[common]),
    lm = unname(o[common]),
    lm_se = unname(res$lm$se[common]),
    abs_delta_vs_lm = unname(abs(b[common] - o[common])),
    within_2_lm_se = unname(abs(b[common] - o[common]) < 2 * res$lm$se[common]),
    row.names = NULL
  )
}, error = function(e) list(error = conditionMessage(e)))

res$pass <- isTRUE(is.data.frame(res$agreement)) && all(res$agreement$within_2_lm_se)
res$generated_on <- as.character(Sys.Date())
saveRDS(res, file.path(RESDIR, "fb-correctness.rds"))
cat("fb-correctness:", if (isTRUE(res$pass)) "PASS" else "see result", "\n")
```

bench/fb-inla-correctness.R

```r
## bench/fb-inla-correctness.R -- real benchmark for the `fb_inla` card.
## Three-way agreement on sleepstudy (lme4), a random-intercept Gaussian LMM:
##   flexyBayes(backend="inla")  vs  standalone INLA::inla()  vs  lme4::lmer().
## lme4 (REML) is the independent oracle; standalone INLA tests flexyBayes's
## translation fidelity (does fb-via-INLA reproduce a hand-written INLA fit?).
```

```r
## Numbers are computed here and cached; the card never hand-types them.
## Run from report/validation/.

RESDIR <- "bench/results"
dir.create(RESDIR, showWarnings = FALSE, recursive = TRUE)

res <- list(card = "fb-inla-correctness", model = "Reaction ~ Days + (1 | Subject)",
            dataset = "sleepstudy (lme4)", oracle = "lme4::lmer REML",
            backend_under_test = "flexyBayes backend='inla'",
            independent_backend = "standalone INLA::inla()")

## load_all FIRST, then seed (load_all consumes the RNG stream).
pkg <- normalizePath(file.path("..", "..", "flexyBayes"), mustWork = FALSE)
ok_pkg <- tryCatch({
  suppressMessages(devtools::load_all(pkg, quiet = TRUE)); TRUE
}, error = function(e) { res$load_error <<- conditionMessage(e); FALSE })
set.seed(20260629L)

utils::data(sleepstudy, package = "lme4")

## -- oracle: lme4 REML ------------------------------------------------------
res$lme4 <- tryCatch({
  t <- system.time(fit_l <- lme4::lmer(Reaction ~ Days + (1 | Subject),
                                       data = sleepstudy, REML = TRUE))[["elapsed"]]
  vc <- as.data.frame(lme4::VarCorr(fit_l))
  list(fixef = lme4::fixef(fit_l),
       se = sqrt(diag(as.matrix(stats::vcov(fit_l)))),
       sd_subject = vc$sdcor[vc$grp == "Subject"],
       sigma = vc$sdcor[vc$grp == "Residual"], seconds = t)
}, error = function(e) list(error = conditionMessage(e)))

## -- independent backend: standalone INLA -----------------------------------
res$inla_standalone <- tryCatch({
  if (!requireNamespace("INLA", quietly = TRUE)) stop("INLA not installed")
  t <- system.time(
    fit_i <- INLA::inla(Reaction ~ Days + f(Subject, model = "iid"),
                        family = "gaussian", data = sleepstudy,
                        control.compute = list(config = FALSE))
  )[["elapsed"]]
  fx <- fit_i$summary.fixed
  list(fixef = stats::setNames(fx[, "mean"], rownames(fx)), seconds = t)
}, error = function(e) list(error = conditionMessage(e)))

## -- backend under test: flexyBayes via INLA --------------------------------
res$fb_inla <- if (!ok_pkg) list(error = "package failed to load") else tryCatch({
  t <- system.time(
    fit_b <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
```

```
                        data = sleepstudy, family = "gaussian", backend = "inla",
                        verbose = FALSE, mcmc_verbose = FALSE)
  )[["elapsed"]]
  list(coef = coef(fit_b), seconds = t,
       class = paste(class(fit_b), collapse = ","))
}, error = function(e) list(error = conditionMessage(e)))

## -- agreement (only when both fb and the oracle produced fixed effects) -----
res$agreement <- tryCatch({
  b <- res$fb_inla$coef; l <- res$lme4$fixef; s <- res$inla_standalone$fixef
  if (is.null(b) || is.null(l)) stop("a required fit did not produce coefficients")
  common <- Reduce(intersect, list(names(b), names(l),
                                   if (!is.null(s)) names(s) else names(l)))
  data.frame(
    term = common,
    fb_inla = unname(b[common]),
    lme4 = unname(l[common]),
    inla_standalone = if (!is.null(s)) unname(s[common]) else NA_real_,
    reml_se = unname(res$lme4$se[common]),
    abs_delta_vs_lme4 = unname(abs(b[common] - l[common])),
    within_1.5_reml_se = unname(abs(b[common] - l[common]) < 1.5 * res$lme4$se[common]),
    row.names = NULL
  )
}, error = function(e) list(error = conditionMessage(e)))

res$pass <- isTRUE(is.data.frame(res$agreement)) &&
  all(res$agreement$within_1.5_reml_se)
res$generated_on <- as.character(Sys.Date())
saveRDS(res, file.path(RESDIR, "fb-inla-correctness.rds"))
cat("fb-inla-correctness:",
    if (isTRUE(res$pass)) "PASS" else "see result (error or out-of-tolerance)", "\n")
```

bench/fb-greta-correctness.R

```
## bench/fb-greta-correctness.R -- real benchmark for the `fb_greta` card.
## The sleepstudy random-intercept LMM via the greta (TensorFlow MCMC) backend:
##   flexyBayes(backend="greta")  vs  lme4::lmer()  (the independent REML oracle).
## This is the SLOW card -- greta initialises TensorFlow and runs HMC, so a full
## run is minutes, not seconds. It is authored runnable; execute with:
##   cd report/validation && Rscript bench/fb-greta-correctness.R
## Numbers are computed and cached; the card never hand-types them. A TF
## non-finite draw (known non-determinism) is recorded, never silently dropped.
## Run from report/validation/.
```

```
RESDIR <- "bench/results"
dir.create(RESDIR, showWarnings = FALSE, recursive = TRUE)
res <- list(card = "fb-greta-correctness",
            model = "Reaction ~ Days + (1 | Subject)",
            dataset = "sleepstudy (lme4)", oracle = "lme4::lmer REML",
            backend_under_test = "flexyBayes backend='greta' (TensorFlow HMC)")

pkg <- normalizePath(file.path("..", "..", "flexyBayes"), mustWork = FALSE)
ok_pkg <- tryCatch({
  suppressMessages(devtools::load_all(pkg, quiet = TRUE)); TRUE
}, error = function(e) { res$load_error <<- conditionMessage(e); FALSE })
set.seed(20260629L)
utils::data(sleepstudy, package = "lme4")

res$lme4 <- tryCatch({
  fit_l <- lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
  list(fixef = lme4::fixef(fit_l),
       se = sqrt(diag(as.matrix(stats::vcov(fit_l)))))
}, error = function(e) list(error = conditionMessage(e)))

res$fb_greta <- if (!ok_pkg) list(error = "package failed to load") else tryCatch({
  t <- system.time(
    fit_b <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                        data = sleepstudy, family = "gaussian", backend = "greta",
                        n_samples = 1000, warmup = 1000, chains = 2,
                        verbose = FALSE, mcmc_verbose = FALSE)
  )[["elapsed"]]
  b <- coef(fit_b)
  list(coef = b, seconds = t, all_finite = all(is.finite(b)),
       class = paste(class(fit_b), collapse = ","))
}, error = function(e) list(error = conditionMessage(e)))

res$agreement <- tryCatch({
  b <- res$fb_greta$coef; l <- res$lme4$fixef
  if (is.null(b) || is.null(l)) stop("a required fit did not produce coefficients")
  if (!all(is.finite(b))) stop("greta/TF produced a non-finite coefficient (TF non-determinism)
  common <- intersect(names(b), names(l))
  data.frame(
    term = common, fb_greta = unname(b[common]), lme4 = unname(l[common]),
    reml_se = unname(res$lme4$se[common]),
    abs_delta_vs_lme4 = unname(abs(b[common] - l[common])),
    within_1.5_reml_se = unname(abs(b[common] - l[common]) < 1.5 * res$lme4$se[common]),
    row.names = NULL
  )
}, error = function(e) list(error = conditionMessage(e)))

res$pass <- isTRUE(is.data.frame(res$agreement)) && all(res$agreement$within_1.5_reml_se)
```

```r
res$generated_on <- as.character(Sys.Date())
saveRDS(res, file.path(RESDIR, "fb-greta-correctness.rds"))
cat("fb-greta-correctness:", if (isTRUE(res$pass)) "PASS" else "see result", "\n")
```

bench/fb-brms-correctness.R

```r
## bench/fb-brms-correctness.R -- real benchmark for the `fb_brms` card.
## The sleepstudy random-intercept LMM via the brms (Stan HMC) backend:
##   flexyBayes(backend="brms")  vs  standalone brms::brm()  vs  lme4::lmer().
## lme4 (REML) is the independent oracle; standalone brms tests flexyBayes's
## translation fidelity. SLOW: brms compiles a Stan model (~30-60s) then runs
## HMC. Authored runnable; numbers cached, never hand-typed. Run from report/validation/.

RESDIR <- "bench/results"
dir.create(RESDIR, showWarnings = FALSE, recursive = TRUE)
res <- list(card = "fb-brms-correctness",
            model = "Reaction ~ Days + (1 | Subject)",
            dataset = "sleepstudy (lme4)", oracle = "lme4::lmer REML",
            backend_under_test = "flexyBayes backend='brms' (Stan HMC)",
            independent_backend = "standalone brms::brm()")

pkg <- normalizePath(file.path("..", "..", "flexyBayes"), mustWork = FALSE)
ok_pkg <- tryCatch({
  suppressMessages(devtools::load_all(pkg, quiet = TRUE)); TRUE
}, error = function(e) { res$load_error <<- conditionMessage(e); FALSE })
set.seed(20260629L)
utils::data(sleepstudy, package = "lme4")

res$lme4 <- tryCatch({
  fit_l <- lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
  list(fixef = lme4::fixef(fit_l), se = sqrt(diag(as.matrix(stats::vcov(fit_l)))))
}, error = function(e) list(error = conditionMessage(e)))

res$brms_standalone <- tryCatch({
  if (!requireNamespace("brms", quietly = TRUE)) stop("brms not installed")
  t <- system.time(
    fit_s <- brms::brm(Reaction ~ Days + (1 | Subject), data = sleepstudy,
                       chains = 2, iter = 2000, refresh = 0, silent = 2,
                       seed = 20260629L)
  )[["elapsed"]]
  fx <- brms::fixef(fit_s)
  list(fixef = stats::setNames(fx[, "Estimate"], rownames(fx)), seconds = t)
}, error = function(e) list(error = conditionMessage(e)))
```

```
res$fb_brms <- if (!ok_pkg) list(error = "package failed to load") else tryCatch({
  t <- system.time(
    fit_b <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                        data = sleepstudy, family = "gaussian", backend = "brms",
                        n_samples = 1000, warmup = 1000, chains = 2,
                        verbose = FALSE, mcmc_verbose = FALSE)
  )[["elapsed"]]
  b <- coef(fit_b)
  list(coef = b, seconds = t, all_finite = all(is.finite(b)),
       class = paste(class(fit_b), collapse = ","))
}, error = function(e) list(error = conditionMessage(e)))

res$agreement <- tryCatch({
  b <- res$fb_brms$coef; l <- res$lme4$fixef; s <- res$brms_standalone$fixef
  if (is.null(b) || is.null(l)) stop("a required fit did not produce coefficients")
  common <- Reduce(intersect, list(names(b), names(l),
                                   if (!is.null(s)) names(s) else names(l)))
  data.frame(
    term = common, fb_brms = unname(b[common]), lme4 = unname(l[common]),
    brms_standalone = if (!is.null(s)) unname(s[common]) else NA_real_,
    reml_se = unname(res$lme4$se[common]),
    abs_delta_vs_lme4 = unname(abs(b[common] - l[common])),
    within_1.5_reml_se = unname(abs(b[common] - l[common]) < 1.5 * res$lme4$se[common]),
    row.names = NULL
  )
}, error = function(e) list(error = conditionMessage(e)))

res$pass <- isTRUE(is.data.frame(res$agreement)) && all(res$agreement$within_1.5_reml_se)
res$generated_on <- as.character(Sys.Date())
saveRDS(res, file.path(RESDIR, "fb-brms-correctness.rds"))
cat("fb-brms-correctness:", if (isTRUE(res$pass)) "PASS" else "see result", "\n")
```

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